

Decoder

2x4 Decoder

w_1	w_0	y_0	y_1	y_2	y_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

Encoder

4x2 Encoder

w_3	w_2	w_1	w_0	y_1	y_0
0	0	0	1	0	0
0	0	1	0	0	1
0	1	0	0	1	0
1	0	0	0	1	1

Priority Encoder

w_3	w_2	w_1	w_0	y_1	y_0	z
0	0	0	0	d	d	0
0	0	0	1	0	0	1
0	0	1	x	0	1	1
0	1	x	x	1	0	1
1	x	x	x	1	1	1

assign $z = !w$

reduction OR

it means $w_0 / w_1 / w_2 / w_3$

use (w)

4'b1xxx: $y = 2'b11$;

4'b01xx: _____;

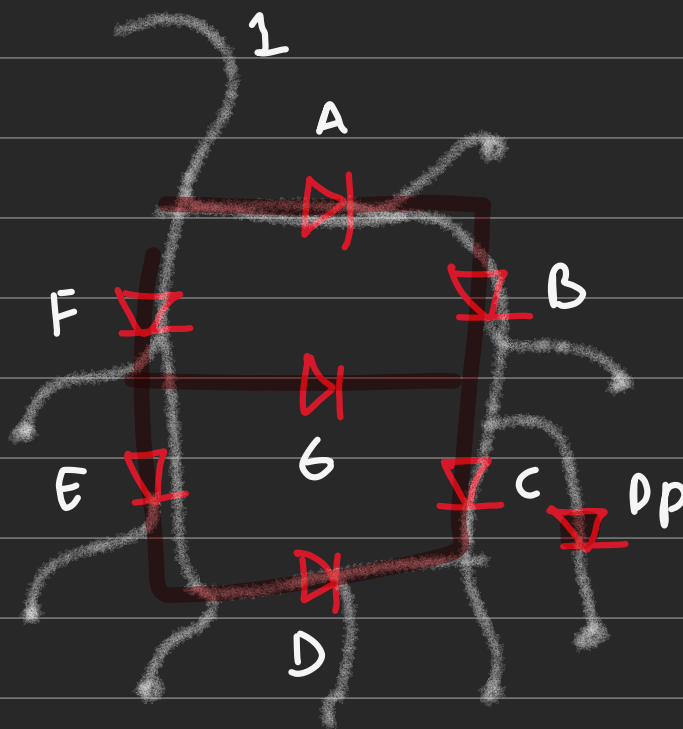
default: $y = 2'bxx$;

end use

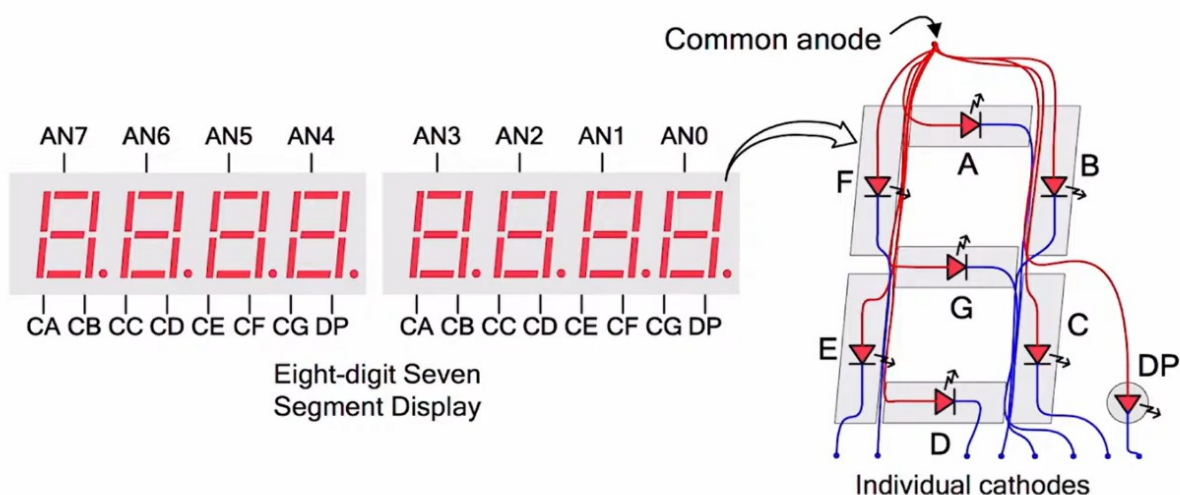
code monitor

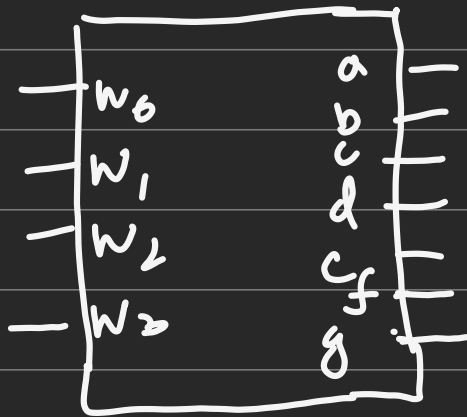
② 7 segment

* common anode



* we give 0 to turn on led
anode $\rightarrow$ 1 0 $\leftarrow$ cathode

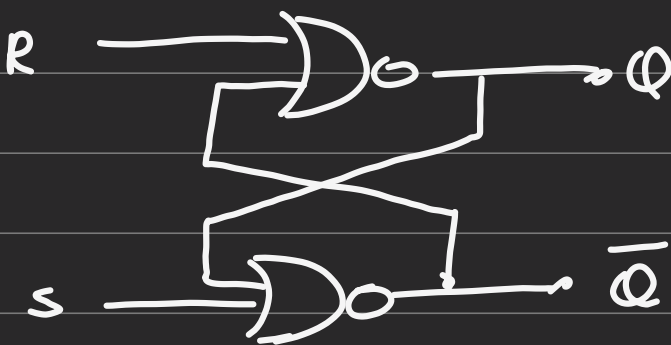




Storage Element

① Simple Storage Elements.

↳ SR Latch

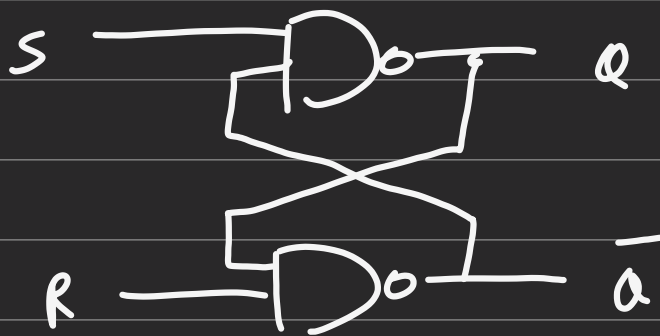


characteristic table

S	R	Q	$\bar{Q}$
0	0	Q	$\bar{Q}$
0	1	0	1
1	0	1	0
1	1	—	—

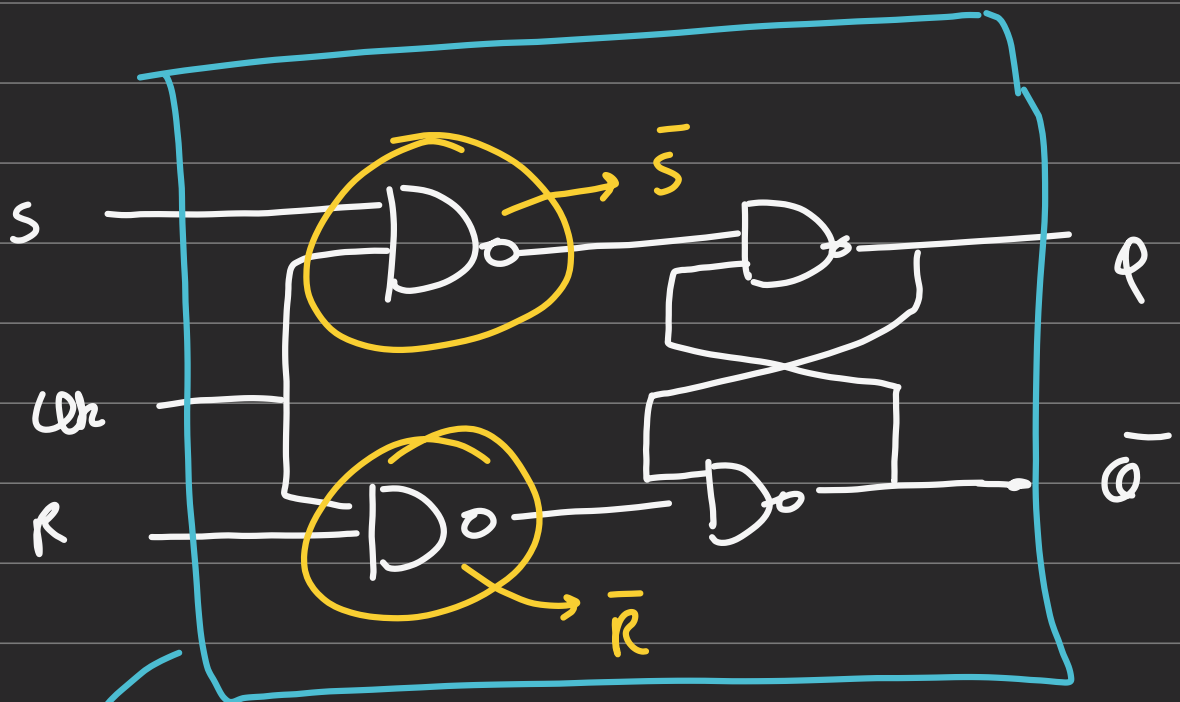
gives
same Q & $\bar{Q}$
which is illegal

* Truth table was
for comb. logic



S	R	Q	$\bar{Q}$
0	0	—	—
0	1	1	0
1	0	0	1
1	1	Q	$\bar{Q}$

↳ Gated SR Latch

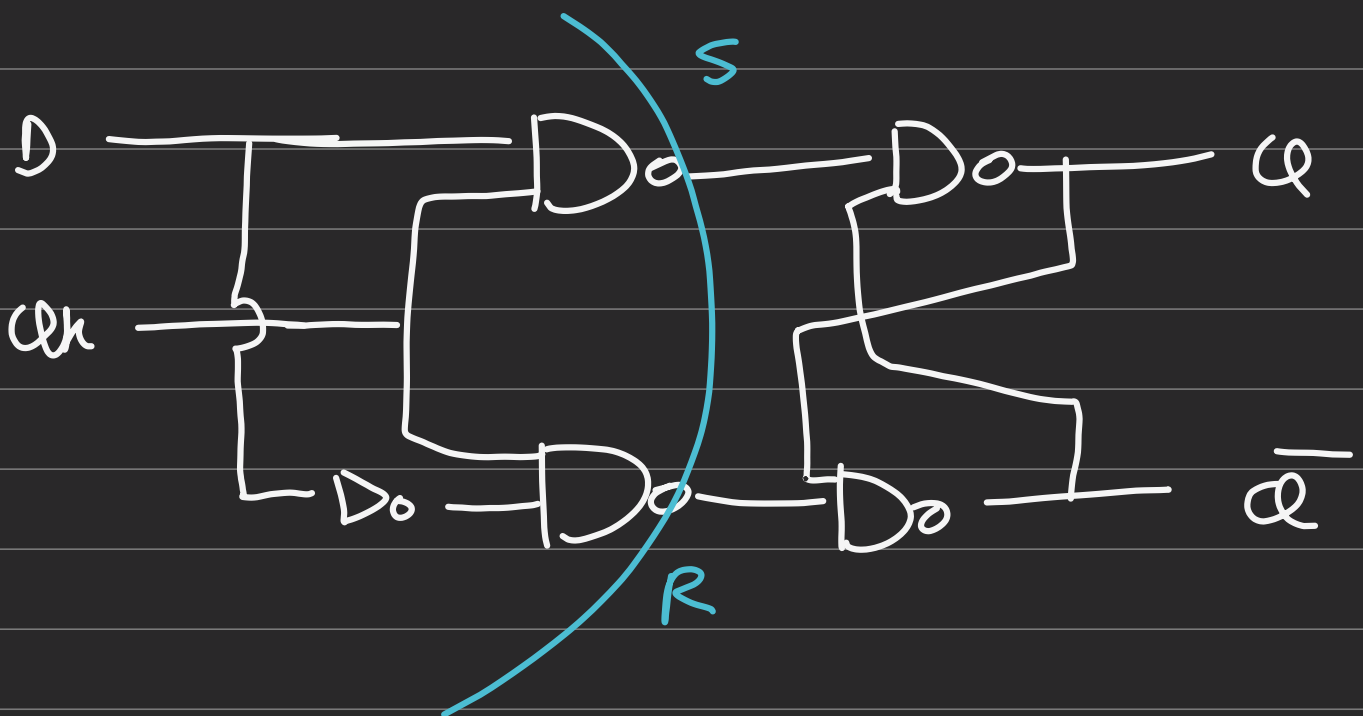


so resulting in equivalent
NOR gate SR latch with
clock

Q	S	R	Q(t+1)
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0	x	x	$Q(t)$
1	0	0	$Q(t)$
1	0	1	0
1	1	0	1
1	1	1	illegal

↳ D Latch



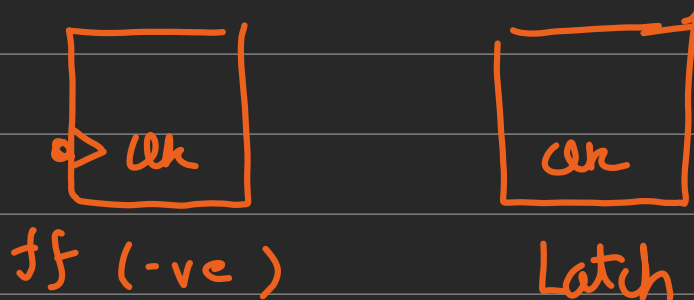
clk	D	$Q(t+1)$
0	x	0
1	0	0
1	1	1

latches $\rightarrow$ level sensitive devices

D Flip-flop



f.f. are edge sensitive devices



$\rightarrow$ generating clock

```
local param T = 20;  
always  
begin  
    clk = 1'b0;
```

(T/2):

clk = 1'b1;

(T/2):

end

→ test bench timing control structure

20

// wait for 20 time units

(2 * T)

// ——— 2 times local parameter T.

@ (negedge rst); // wait for rst to deassert.

@ (negedge clk); // ——— neg clk edge

repeat (10) @ (negedge clk); // wait for 10 negative clk edges

wait (x == 2); // wait for x to become 2.

wait (y)

// ——— y to —
— 1 .

\$ stop.

// stop simulation

